452. Richard Wolffenstein: On the Action of Hydrogen Peroxide on Acetone and Mesityl Oxide.

(Received August 15.)

English translation of Ber. dtsch. chem. Ges. 28 (1895) 2265–2269. Structural formulas are the originals. This is the paper in which tricycloacetone peroxide (acetone peroxide) was first described.

In the action of hydrogen peroxide on coniine[1]) I used acetone as the most suitable solvent for coniine in the hydrogen peroxide solution, and expressly emphasized that the acetone was here not attacked by the hydrogen peroxide. That was indeed fully the case with the dilute (one-percent) hydrogen peroxide solution used there. It is otherwise, however, with concentrated solutions.

Thus, from a solution of equal amounts of acetone and 10-percent hydrogen peroxide, after standing several days at ordinary temperature, small amounts of crystals separated. These were filtered off; after a few days a further crystal deposit had again formed in the solution. This experiment could be repeated several more times with the same result. Yet the crystal amount collected was so sparse that it almost seemed as if these crystals owed their origin to the acetaldehyde scarcely removable from the acetone. But a direct experiment made to test this came out negative.

The excellent crystallizing power of the new compound did, on the other hand, considerably facilitate its preparation in the pure state, and so, after weeks of effort, at least an amount sufficient for the analyses was collected. For analysis they were recrystallized from ether. M.p. 97° C.

Analysis: Calc. for $C_3H_6O_2$.

Percent: C 48.7, H 8.1. Found: C 48.6, 48.2; H 7.9, 8.3.

There was accordingly present a compound of the composition of an acetone peroxide, $(H_3C)_2C<\begin{matrix}O\\|\\O\end{matrix}$.

The crystals, of neutral reaction, are insoluble in water, acids and alkalis, readily soluble in benzene, acetone, ether etc. Heated on a platinum spatula, they deflagrate. They possess a spicy odor.

On warming, indeed even on standing in air, they volatilize somewhat: 0.1820 g of substance, spread on a watch glass, lost after 48 hours 0.0059 g.

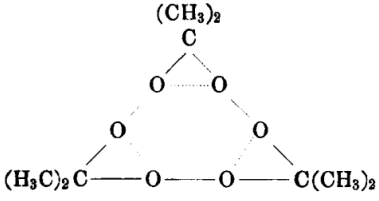
In a desiccator over concentrated sulfuric acid a formal absorption even takes place, similar to what acetone also shows: 0.2052 g lost after 48 hours 0.1016 g.

The physical properties of the peroxide — the solid state of aggregation, the insolubility in water etc. — argued that its molecular weight is greater than corresponds to the simple atomic ratio. The molecular weight was therefore determined from the freezing-point depression of benzene.

0.1949 g of substance dissolved in 17.186 g of benzene lowered the freezing point by 0.27° C.

Calc. for $[C_3H_6O_2]_3 = 222$. Found = 206.

There was thus present a trimolecular acetone peroxide, which can arise from the monomolecular one by the bonds between each two oxygen atoms breaking and serving for linkage with the oxygen atoms of a neighboring molecule. One thus arrives at the following constitutional formula:



This peculiar ring-constituted compound shall be called **tri-cycloacetone peroxide**.

After it had been recognized by the analysis that it had formed from molecular amounts of acetone and hydrogen peroxide, this proportion was later also applied in the preparation, and one finally reached nearly quantitative yields. Thus, for example, 11.6 g of acetone were mixed with 13.6 g of 50-percent hydrogen peroxide under cooling. After four weeks' standing, 11.9 g of analytically pure crystals had separated. Especially striking in the formation of this cycloacetone peroxide was that it did not form from pure acetone and pure distilled hydrogen peroxide[1]), but only on the addition of a drop of phosphoric acid; whereby the formation of this peroxide appears at the same time as a

condensation process, which also agrees with the constitutional formula.

Above all, however, it was confirmed by the chemical behavior of the compound. As was to be expected from the ring structure and the indifferent atom groups composing it, the compound proved little reactive. Of the many experiments in this regard I will emphasize only the complete lack of reaction toward acetic anhydride, since this excludes the presence of an oxygen atom as hydroxyl oxygen. On the other hand, the acetone peroxide showed its peroxide structure in that, on warming for about an hour with dilute sulfuric acid on the water bath, it decomposed completely again into acetone and hydrogen peroxide. The tri-cycloacetone peroxide, wholly insoluble in water, had then disappeared, and in the resulting solution both the hydrogen peroxide could be readily detected by the various reagents, and the acetone by the iodoform precipitate on warming with iodine in alkaline solution:

$$[C_3H_6O_2]_3 + 3\ H_2O = 3\ C_3H_6O + 3\ H_2O_2.$$

Its peroxide nature also manifested itself in another remarkable property: in its enormous explosiveness, which makes great caution advisable when working with the acetone peroxide. Without external cause I have, it is true, never observed an explosion, but on impact and shock, or on heating, the peroxide exploded with the greatest brisance. Thus once, during an elementary analysis already underway for some time, the tube exploded so violently that the part of the tube where combustion was just proceeding was shattered to finest glass powder, while the end and the front part of the combustion tube were displaced by only about one cm. When, on another occasion, the acetone peroxide was to be sealed into a shooting tube, a tiny crystal of the compound had probably remained hanging on the tube wall, which on heating before the blast flame gave rise to such an explosion that the tube was flung from the hand[1]).

The reaction that acetone underwent with concentrated hydrogen peroxide prompted the investigation of mesityl oxide in the same sense. Here, however, the action proceeds in a different direction.

If, for example, 10 g of mesityl oxide were treated with 100 g of 10-percent hydrogen peroxide — this proportion proved the most favorable — the mesityl oxide, at first floating on the surface,

gradually took on a consistent nature and sank. The reaction took place at ordinary temperature and was promoted by frequent shaking. After several months the action of the hydrogen peroxide was finished; it was filtered off, the reaction product first cleaned on clay plates, then recrystallized from ether-alcohol. Thus one obtained dazzling white granular crystals; insoluble in water, dilute acids or alkalis, readily soluble on the other hand in acetone, benzene, ether etc. The reaction of the new compound was neutral. M.p. 123° C. without decomposition.

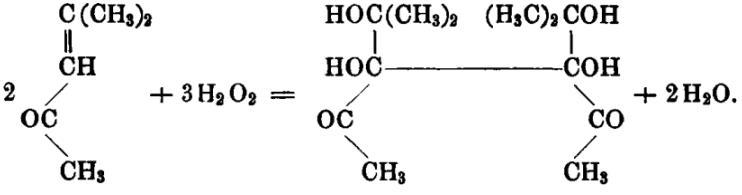
Analysis: Calc. for $C_6H_{11}O_3$.

Percent: C 54.9, H 8.4. Found: C 54.5, 54.5; H 8.3, 8.6.

The determination of the molecular weight gave the following: 0.1997 g dissolved in 14.1102 g of benzene gave a freezing-point depression of 0.275°.

Calc. for $[C_6H_{11}O_3]_2$ 262. Found 239.

In mesityl oxide, then, there occurs not, as in acetone, an addition of one oxygen atom through the action of hydrogen peroxide, but an addition of one hydrogen-peroxide molecule (with simultaneous exit of one hydrogen atom) according to the following formula equation:



The breaking of the double bond between two carbon atoms by addition of hydrogen peroxide is not without analogy. Thus the glycols form from the olefins on the action of hydrogen peroxide[1]).

This hyperoxide shows its chemical character in its action on hydroxylamine, which is thereby oxidized to nitric acid; with phenylhydrazine, after the reaction is once initiated by warming, a regular stream of nitrogen develops, while the phenylhydrazine passes into benzene — reactions which belong to hydrogen peroxide itself[2]). On warming with dilute sulfuric acid on the water bath, the hyperoxide decomposed again into hydrogen peroxide and mesityl oxide.

The pronounced explosive power that the tri-cycloacetone peroxide possesses, this hyperoxide does not have. On striking it with a hammer, only a roughly whip-crack-like detonation occurs; in the analysis, however, special care must be taken to ensure a slow course of combustion.

Berlin. Organic Laboratory of the Technical University.

---

[1]) (coniine) Diese Berichte 28, 1459. (pure H_2O_2) [1]) Diese Berichte 27, 3307.

(explosion) [1]) The explosiveness that hydrogen peroxide sometimes shows from »unknown causes« evidently rests on the decomposition of such labile peroxides, especially of ether, which is often used for purifying hydrogen peroxide. (glycols) [1]) Carius, Ann. d. Chem. 126, 209. [2]) Wurster, diese Berichte 20, 2631.